

ASCII MASTER FLOW – PANEL 1/12: Geological control of resource distribution

GEOLOGICAL HISTORY -> rock structure, basins, tectonics, surface weathering

SHIELD / IGNEOUS-METAMORPHIC TERRAIN -> metallic ores

SEDIMENTARY BASIN -> coal, petroleum, natural gas

TROPICAL LATERITISATION -> bauxite on suitable plateaux and hilltops

TRAP -> distribution is a deduction from Earth history, not a country list

MUST REMEMBER: Resource geography begins with geological occurrence, reserve/resource classification, grade, extraction technology, transport, processing, markets and externalities; world belts and India's Gondwana coal, offshore/onshore hydrocarbons, Odisha-Jharkhand iron-manganese-bauxite, Rajasthan non-ferrous minerals and nuclear/renewable endowments require map logic.

ASCII MASTER FLOW – PANEL 2/12: Mineral and energy classification ladder

FERROUS METALLIC -> iron ore, manganese, chromite; steel-making base

NON-FERROUS METALLIC -> copper, bauxite, lead, zinc

NON-METALLIC INDUSTRIAL -> mica, limestone, gypsum, phosphates

MINERAL FUELS -> coal, petroleum, natural gas, uranium

ENERGY PAIRS -> conventional vs non-conventional; exhaustible stock vs renewable flow

ASCII MASTER FLOW – PANEL 3/12: World mineral-energy belt rail

PERSIAN GULF -> classic sedimentary hydrocarbon province

APPALACHIAN / INTERIOR USA -> coal, oil, gas with iron associations

RUHR-LORRAINE -> historic coal-iron heavy-industry core

RUSSIAN-SIBERIAN / CHINESE INTERIOR -> continental-scale fuel frontiers

COPPERBELT, AUSTRALIAN SHIELD, ANDES -> shield and tectonic metal provinces

ASCII MASTER FLOW – PANEL 4/12: Extraction to industry chain

EXTRACTION -> ore or fuel raised at the deposit

TRANSPORT -> rail, slurry pipeline, port or inland waterway evacuation

PROCESSING -> beneficiation, smelting, refining, petrochemical conversion

POWER + MARKET -> dispatchable energy, capital and demand complete the cluster

RULE -> a producing region is not automatically an industrial region

ASCII MASTER FLOW – PANEL 5/12: Occurrence to usable resource filter

OCCURRENCE EXISTS -> geology has done its part
IF grade and depth are workable -> continue to the economic test
IF transport distance and technology permit -> continue to the price test
IF price sustains the cost -> occurrence becomes a usable resource
ELSE -> the deposit stays a geological fact with no economic geography

ASCII MASTER FLOW – PANEL 6/12: Four-element petroleum system

SOURCE ROCK -> organic-rich fine sediment buried deeply and long enough

RESERVOIR ROCK -> porous permeable sandstone or fractured carbonate

SEAL / CAP ROCK -> shale, evaporite or tight limestone preventing escape

TRAP -> anticline, fault block, salt dome, reef or stratigraphic pinch-out

VERDICT -> sedimentary basin is necessary but never sufficient

ASCII MASTER FLOW – PANEL 7/12: Why passive margins hold offshore oil

SHELF AND SLOPE -> the world's great sediment sinks receive river-borne organic load

OXYGEN-POOR BOTTOM WATER -> marine settings preserve organic matter from decay

PASSIVE MARGIN SUBSIDENCE -> thick, tectonically quiet sediment prisms accumulate

DELTA AND SALT TECTONICS -> pair source with reservoir and generate trap families

PRESERVATION EDGE -> uplift and erosion have breached many onshore accumulations

ASCII MASTER FLOW – PANEL 8/12: Offshore versus onshore comparison

GEOLOGY -> onshore intracratonic and rift basins vs shelf, slope and deepwater margin

COST AND TECHNOLOGY -> incremental land drilling vs lumpy high-cost marine systems

THRESHOLD -> small onshore fields can be viable; offshore needs large fields

FOOTPRINT -> land acquisition onshore vs fisheries, shipping and spill risk offshore

JURISDICTION -> ordinary national law vs maritime zones and contested boundaries

ASCII MASTER FLOW – PANEL 9/12: India petroleum province map hooks

ONSHORE -> Assam-Arakan basin and the Cambay basin of western India

WESTERN OFFSHORE -> major continental-margin province, carbonate reservoir character

EAST-COAST OFFSHORE -> Krishna-Godavari and Cauvery deltaic margins, gas-prone

COASTAL CONSEQUENCE -> landfall terminals pull refining and petrochemicals to the coast

CAUTION -> name provinces and reservoir character, never remembered output figures

ASCII MASTER FLOW – PANEL 10/12: Critical-mineral chokepoint fork

TRANSITION DEMAND RISES -> lithium, cobalt, nickel, graphite, copper, rare earths

IF the question is extraction -> supply is geologically narrow but visible

IF the question is processing -> concentration is tighter and often in other countries

THEREFORE -> the binding chokepoint is midstream, not at the mine

RESPONSE -> recycling, substitution, diversification and strategic stockpiling

CLOSE DISTINCTION: Resource differs from reserve, occurrence from economically recoverable deposit, thermal from metallurgical coal, conventional from non-conventional only by stated taxonomy, and installed renewable capacity from actual generation; criticality combines economic importance and supply risk rather than geological scarcity alone.

ASCII MASTER FLOW – PANEL 11/12: India mineral belt and energy corridor rail

CHOTA NAGPUR COMPLEX -> iron ore, coal, mica, bauxite, manganese heartland

BELT HOOKS -> Bailadila iron, Jharia coking coal, Khetri copper, Zawar lead-zinc

GONDWANA COAL -> Damodar, Mahanadi, Son and Godavari valley basins

SOLAR CORRIDOR -> Rajasthan and Gujarat dry high-insolation belts

WIND CORRIDOR -> Tamil Nadu, Gujarat, Karnataka and Maharashtra coasts and passes

EVIDENCE LIMIT: Jharia/Raniganj/Talcher/Korba, Mumbai High, Assam, Krishna-Godavari,

Bailadila, Keonjhar, Khetri, Jaduguda, monazite coasts, solar Rajasthan/Gujarat and wind

Tamil Nadu/Gujarat anchor India; Ministry, IBM, CEA and international data need year/status

labels, while just-transition and import dependence qualify security claims.

ASCII MASTER FLOW – PANEL 12/12: Mineral and energy answer spine

DEFINE -> which class of resource and which measure is being asked about

DEDUCE -> shield, basin or weathering control before naming any place

LOCATE -> world belt, then the Indian belt or petroleum province

QUALIFY -> occurrence versus usable resource; capacity versus generation

CLOSE -> midstream chokepoint, institutional conversion and the dated mission anchor